

Part 2: Mitigation, Market Sizing, Cost to Capture & Realistic Payback

Poultry and cold storage, stress-tested against realistic assumptions

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Continues directly from Part 1 (SME Market & Financial Case).

Covers Sprints A–D: mitigation, TAM/SAM, cost to capture, realistic capture & payback.

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Executive Summary

Part 1 found that a plain “solar savings beat your EMI” pitch fails for both poultry and cold storage. This part asks the natural next questions: can it be fixed, how big is the prize, what does capturing it cost, and what does realistic success actually look like.

Four findings, in order

1) Yes, it can be fixed — mostly through financing, not solar engineering. Bangladesh Bank's own Green Refinance Scheme (3–5% vs. the 11–13% commercial rate modeled in Part 1) is the single biggest lever, and it already exists. **2)** The two-vertical TAM is Tk50–63B (~\$410–513M) — a believable ~20–25% slice of Bangladesh's total solar market. **3)** Reaching a credible first milestone (1,500 poultry farms + 15 cold storage facilities) costs SolarThik itself about Tk33M (~\$270K) over 24 months — operating cost, not the hardware capital, which flows through banks. **4)** Under realistic (“base-case”) growth assumptions, these two verticals alone do **not** clearly pay back within 10 years — they come close but don't cross. Only an optimistic capture scenario clearly works. This is a proof-of-concept-scale business on its own, not the whole company.

Part C — Mitigating the Unit-Economics Gap

C1. The core lever: cheap capital that already exists

Bangladesh Bank runs a Green Refinance Scheme capping the customer-facing rate at **5% (as low as 3% for solar linked to agriculture)**, tenures up to 10 years, through 44 banks and 25 financial institutions already signed up. IEEFA's own research describes utilization of this fund as “stubbornly low” due to documentation friction, awareness gaps, and slow disbursement — precisely the problem a feasibility-and-application concierge service is built to solve. This is not a capital source we need to invent or lobby for. It is sitting there, underused, today.

C2. Poultry mitigation stack (2,000-bird shed)

Scenario	EMI/month	Solar savings/month	Net position
Baseline (11% commercial loan)	Tk 8,656	Tk 4,819	–Tk 3,837
+ Green refinance only (3%, 8yr)	Tk 5,648	Tk 4,819	–Tk 829
+ EC fan retrofit + right-sized critical-load battery	Tk 4,487	Tk 4,819	+Tk 332

EC fans also cut the base bill by ~Tk5,300/month independent of solar entirely (sourced payback: 12–24 months from energy savings alone), and right-sizing the battery to just the ventilation circuit — not the whole shed — roughly halves battery cost while still protecting the actual life-or-death load.

C3. Cold storage mitigation stack (3,000-tonne facility)

Scenario	EMI/month	Value/month	Net position
Baseline (12% commercial, solar only)	Tk 120,039	Tk 100,602	–Tk 19,436
+ Green refinance only (5%, 10yr)	Tk 76,632	Tk 100,602	+Tk 23,970
+ VFD retrofit + peak-shaving battery	Tk 76,632	Tk 233,862	+Tk 157,230

Surprising finding

VFD retrofit alone — independent of solar — cuts the bill by roughly Tk418,000/month, more than the entire solar system's own value. For cold storage specifically, efficiency-plus-financing may be the lead product, with solar as the second layer. Caveat: sourced savings (10–35%) are solid; no reliable Bangladesh-specific VFD capex figure was found, so this is not yet fully costed.

Part D — Market Sizing (TAM/SAM)

D1. Poultry

Tier	Farms	System value
TAM — all 100,000 commercial farms	100,000	Tk 41.0–53.5B (\$336–439M)
SAM — grid-tied, bankable scale (est. 55% of TAM*)	~55,000	Tk 22.6–29.4B (\$185–241M)
SAM, Dhaka division only	~29,300	Tk 12.0B (\$99M)
SAM, Gazipur district only	~13,200	Tk 5.4B (\$44M)

*The 55% filter is a reasoned estimate, not a sourced figure — no Bangladesh-specific farm-size distribution dataset exists to pin this down precisely.

D2. Cold storage — including a data correction

Correction from Part 1

Part 1 used an older source citing 400 facilities / 6M tonnes capacity. A more recent (May 2025) report puts the figure at **365 facilities / 3.2M tonnes** — used here as the primary figure. Both are shown for transparency; the newer, more granular source (with district-level breakdowns) is preferred.

Tier	Facilities	System value
TAM — all 365 facilities, solar-only	365	Tk 9.1B (~\$74M)
SAM (excl. ~15% assumed too distressed to finance)	~310	Tk 7.7B (~\$63M)

VFD/efficiency-retrofit value is deliberately excluded from this TAM — the savings are sourced, the capex is not, so it is not included until real vendor quotes exist.

Combined two-segment TAM: Tk 50–63B (~\$410–513M)

Roughly 20–25% of Bangladesh's entire estimated solar market (\$2.1B, all segments) — a believable slice for two specific verticals, not an inflated one. **This is total system value, not SolarThik's own revenue** — our revenue is a thin origination-and-monitoring-fee layer on top, sized properly in Part F.

Part E — Cost to Capture the Market

E1. Core team, Year 1 (monthly cost)

Role	Cost/month (Tk)
Regulatory/compliance lead (SREDA, BERC, net-metering)	70,000
BD/partnerships lead (integrators, associations, banks)	60,000
Credit/risk analyst	80,000
Software engineers × 2 (platform/monitoring)	100,000
Field ops / installer-QA coordinator	40,000
Field agents/site assessors × 3	90,000
Founder/ops overhead	60,000
Total	500,000/month (~Tk 6M, ~\$49K/yr)

Bangladesh salary benchmarks online were inconsistent across sources (one aggregator showed an obviously mislabeled “Tk55,000/year” software-engineer figure); the more internally-consistent monthly figures were used instead. Treat as planning-grade.

E2. Customer acquisition cost by channel

- **Poultry, integrator-channel:** one relationship deal (~Tk400,000) unlocks ~250 farms → blended CAC ~Tk7,600/farm.
- **Poultry, independent farmer:** field-agent-driven, higher touch → ~Tk22,000/farm — nearly 3x the integrator-channel cost. This is the concrete payoff of channeling through integrators.
- **Cold storage:** ~Tk350,000/facility — about 1.4% of the average deal's system value, normal for high-ticket B2B (1–5% typical).

E3. Cost to reach a first credible milestone (24 months)

Target: 1,500 poultry farms + 15 cold storage facilities	Cost
Poultry CAC (1,200 integrator + 300 independent)	Tk 15.7M
Cold storage CAC (15 facilities)	Tk 5.25M
2-year team cost	Tk 12.0M
Total	Tk 33.0M (~\$270,000)

What this number is (and isn't)

This is SolarThik's own operating capital to acquire and serve customers — not the Tk8.5–24.8M per-system hardware/financing cost, which flows through banks/NBFI/IDCOL refinance under the packager model. \$270K to a real proof point is within reach of Bangladesh's own new bank-backed startup fund (Tk4.25B, formed Dec 2025), without needing foreign VC on day one.

Part F — Realistic Capture Rate & Company Payback

F1. Methodology — including a transparent correction

Built-in error, caught and fixed before delivery

The first version of this model flagged “payback in Year 1” for every scenario. That was a bug: it detected the first time cumulative cash flow crossed zero and never checked whether it stayed there. An early pilot-year fee bump created a false positive while later years went deeply negative. The model was rebuilt to require **sustained** positive cumulative cash flow through Year 10, not just a first crossing, and rerun in full below. Flagging this explicitly because the brief was “do not hallucinate or give vague work,” and silently shipping the buggy number would have violated that directly.

Growth-curve shape: an S-curve (logistic), not linear — informed by SolarSquare's real trajectory (~10 years to ~50,000 households, with the visible acceleration concentrated in the final ~18 months). Revenue = a 4% origination fee on new system value + a recurring monitoring fee (Tk250/mo poultry, Tk12,000/mo cold storage, reflecting the larger dispatch-optimization value created there) on all active systems. Team cost scales in three tiers (Year 1–2 prove-it, Year 3–5 scale-up, Year 6–10 mature) rather than staying flat.

F2. Three-scenario results

Scenario	Trough (worst point)	Sustained payback	Yr-10 customers	Yr-10 cum. revenue
Conservative	–Tk151M (Yr10, still falling)	Not reached in 10yr	4,614 poultry / 39 CS	Tk0.22B (~\$1.8M)
Base	–Tk13M (Yr7), recovering	Not reached in 10yr	8,977 poultry / 79 CS	Tk0.43B (~\$3.5M)
Optimistic	+Tk5M (Yr3, barely dips)	Year 1	14,670 poultry / 118 CS	Tk0.69B (~\$5.6M)

Capture rates behind these figures: conservative reaches ~8% of poultry SAM / ~13% of cold storage SAM by Year 10; base reaches ~16%/25%; optimistic ~27%/38% — all well below fantasy full-market capture, and broadly consistent with how much of its own home market SolarSquare has actually reached at real scale.

F3. Strategic implication

The honest bottom line of this entire exercise

Under the **base case** — the scenario grounded in a real comparable's actual growth curve — poultry and cold storage **do not clearly pay back within 10 years**. The deficit shrinks from –Tk13M (Year 7) to just –Tk6M (Year 10) and is visibly still improving, but doesn't cross zero in the modeled window. Only the optimistic case clearly works. These two verticals top out around \$3.5–5.6M in cumulative 10-year revenue in the realistic-to-optimistic range — a solid, fundable proof of concept, but not a venture-scale outcome by themselves.

This connects directly to the earlier billion-dollar strategy memo's insistence on three things that were flagged there as important and are now shown here to be **structurally necessary, not optional**: keep residential/commercial in the product (Part 1, Section B3 — the one segment where bill-savings alone already works), pursue additional SME verticals once properly researched (the ETP/textile lead flagged in Part 1, Section A3), and treat regional expansion as core strategy, not a stretch goal. Poultry and cold storage are the credible beachhead that builds the lending relationships and the repayment/generation data — the data moat described in the billion-dollar memo. The venture-scale outcome has to come from stacking more verticals and more geography on top of this proven playbook, not from squeezing more share out of 55,000 poultry farms and 310 cold storage facilities alone.

Bottom Line

- **Mitigation works** — mostly through routing customers into Bangladesh Bank's already-existing, under-used Green Refinance Scheme (3–5%), not through solar engineering changes.
- **The market is real and sizeable**: Tk50–63B (~\$410–513M) combined TAM across poultry and cold storage — but that is total system value flowing through the market, not SolarThik's own revenue.
- **Reaching proof-of-concept scale is cheap**: ~Tk33M (~\$270K) in operating capital, achievable from Bangladesh's own new bank-backed startup fund without foreign VC.
- **Realistic payback on these two verticals alone is not clearly positive within 10 years** — the base case comes close but doesn't cross. Treat poultry and cold storage as the beachhead, not the whole company.

Methodology & Sources

Financing: Bangladesh Bank Green Refinance Scheme circulars and IEEFA analysis of utilization rates. Efficiency: Volcano Electric (EC fan sizing/payback), Rinac and multiple VFD/refrigeration industry sources (savings ranges; no BD-specific capex source found). Market sizing: earlier-cited poultry sector sources (Part 1) plus an updated cold-storage source (The Business Standard, May 2025) which supersedes the Part 1 figure, both shown for transparency. Salary data: BDTechJobs, Payscale, and Levels.fyi cross-checked against inconsistent Glassdoor figures, with the more internally-consistent monthly figures used. Growth-curve reference: SolarSquare (India), per TechCrunch/Tracxn reporting cited in the earlier billion-dollar strategy memo. All TAM, SAM, CAC, team-cost, and capture-rate figures are original modeling built from these inputs, not directly quoted third-party numbers — flagged as such throughout. Nothing in this document is a loan offer, financial advice, or an engineering specification; every figure needs verification against real quotes, audits, and term sheets before customer- or investor-facing use.